

Arvind Prabhu S

✉ arvindprabhus15@gmail.com ☎ 9363130512 📍 Chennai 🔗 [linkedin.com/in/arvind-prabhu-s-6bb869290](https://www.linkedin.com/in/arvind-prabhu-s-6bb869290)
🌐 <https://github.com/Vindi77>

Profile

ECE undergraduate with hands-on experience in Embedded C, STM32, ESP32 and PCB design. Strong interest in firmware development, peripheral interfacing, and real-time systems.

Professional Experience

Sri SivaSubramaniya Nadar (SSN) College of Engineering – Chennai, India, Intern 05/2025 – 06/2025
Chennai
- Designed and fabricated custom single-layer PCBs for an ESP32-based classroom automation system, integrating PIR sensors and relay driver circuits.
- Contributed to a research project on ESP32-based automation systems

Projects

Task Scheduler using SysTick

- Round-robin scheduler on STM32 Nucleo with LED task demo.
- Used bare-metal C & ARM assembly with custom task stacks and GPIO handling.

STM32 Bare Metal Driver Development

- Wrote register-level drivers for GPIO, UART, I2C, SPI with interrupt handling
- Focused on direct CMSIS programming

IoT-Based Smart Classroom Automation System

- Designed and fabricated a custom single-layer PCB for an ESP32-based automation system

Wireless Control System using NRF Modules

- Arduino Nano + NRF24L01 based wireless TX-RX control system
- Controlled motors and servos via custom PCB with real-time actuation

Skills

Programming

C, Python, ARM Assembly

Embedded Software

GPIO, UART, SPI, I2C, Bare-Metal programming, CMSIS

Tools and Platforms

STM32CubeIDE, KiCad, EasyEDA, LTspice, Git

Fundamentals

Analog & Digital Electronics, Signal Processing

Achievements

Winner – Synthetix 3.0 Circuit Design Hackathon, 01/05/2025
VIT Chennai, 30 hour national-level hackathon

Education

New Horizon Public School, 10th ICSE Bangalore
97%

Vidya Mandir Senior Secondary School, 12th CBSE Chennai
96.2%

SRM Easwari Engineering College, BE Electronics and Communication Engineering Chennai
Pre-final Year student CGPA : 8.96
2026 - Present